

JONATHAN EARP

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I've worked on StoveCam at Lawrence Berkeley National Laboratory since 2024, from building the sensor to grading its output by hand against 506 captures. In summer 2025 I did nanofabrication in the Class 100 cleanroom at UPenn's Singh Center.

EDUCATION

University of Edinburgh Edinburgh, UK · 2026 – 2031
MEng Electrical and Mechanical Engineering, Year 1

The Head-Royce School Oakland, CA · Jun 2026
GPA 3.80 unweighted, 4.38 weighted. Honors Calculus, Advanced Physics, Advanced Chemistry, Data Science

EXPERIENCE

Lawrence Berkeley National Laboratory — Student Research Assistant Berkeley, CA · Apr 2024 – Present
StoveCam: low-cost camera sensor for an indoor air quality and childhood asthma study

- Built a Raspberry Pi 5 unit with RGB and thermal cameras that captures the cooktop every 60 s in occupied homes, then reworked the mounts, wiring and capture scripts until it ran reliably across different kitchens.
- Wrote a three-stage Python labeling pipeline: an on-device brightness gate, a low-cost cooking screen, then Gemini vision for food, method, burner and cookware. 40,000+ RGB/thermal pairs from three homes at \$0.018 per analyzed pair.
- Validated the output against 506 hand-graded pairs: mean score 8.1/10, with zero burner or cooking-method errors. Most failures traced to residual-heat false positives. Abstract submitted to ISES 2026.

University of Pennsylvania, Singh Center for Nanotechnology — ESAP Participant Philadelphia, PA · Summer 2025

- Class 100 (ISO 5) cleanroom: photolithography, microfluidics, CdSe quantum dot synthesis, SEM and AFM imaging.
- Built an adaptive audio control device with Prof. Gyuseok Kim and graduate TAs, from first prototype to working demo.

PROJECTS

3D Reconstruction from FPV Flight Data — S.P.L.O.O.G.E | *Python, COLMAP, Rust, WebGL* Aug 2026

- Built a pipeline from raw DJI O4 Pro video and gyro telemetry to one 3D Gaussian Splatting model of two flights six days apart: 7,470 of 7,471 frames registered at 0.82 px mean reprojection error.
- Estimated the uncalibrated lens focal length by regressing optical flow against gyro rotation ($R^2 = 0.967$), and cut video-to-render misalignment from 72.3 px to 0.89 px by undistorting the footage.
- Confirmed 13,482 verified image matches linked the two sites before committing to an 18-hour mapping run.
- Traced a rendering defect to the viewer's near-plane cull after measuring four other fixes, including a depth-distortion loss I implemented in Rust. Holes fell from 51.2% to 4.4%, with no regressions across 56 test poses.

FPV and UAS Builds — Personal | *Betaflight, ArduPilot, ExpressLRS* 2023 – Present

Head-Royce VEX Robotics — Team Captain Oakland, CA · 2022 – 2026

Automated Video Rendering Pipeline — P.E.T.E.R | *Python, OpenCV, Playwright, ffmpeg* 2026

SKILLS

Programming: Python (NumPy, OpenCV, Playwright), Rust, COLMAP, JavaScript, WebGL, Arduino, Raspberry Pi / Linux, Git

Controls & sensors: PID tuning, IMU and gyro data, quaternion orientation, sensor calibration, flight-log analysis

CAD & fabrication: Onshape, Fusion 360, 3D printing, CNC, laser cutting, bandsaw, drill press, photolithography

Electrical & avionics: Brushless motors, STM32 ESCs, soldering, Betaflight, ArduPilot, ExpressLRS, analog and DJI video

LEADERSHIP

- Medical Advocacy, President.** Ran a blood drive that brought in 120 donors, and set up a partnership with a local hospital.
- Eagle Scout, Senior Patrol Leader.** Taught a month-long robotics class at an elementary school for my Eagle project, and left the curriculum behind so someone else could run it again.